TECHNICAL SPECIFICATION FOR VERTIAL LIFTING FABRIC DOOR

Scope of Supply

Design, Supply, installation, testing, and commissioning of electrically operated, vertically lifting fabric doors (hoist-up door system), designed and manufactured by a technically qualified and experienced entity. The door system shall be constructed in accordance with internationally recognized performance standards, including but not limited to EN 13241 latest standard, and shall utilize high-strength, industrial-grade fabric materials. The system must meet all relevant performance, safety, durability, and operational requirements as detailed in the technical specifications.

The complete door assembly shall be designed to provide effective weather sealing on both the interior and exterior faces. The bottom section of the door assembly shall be engineered to ensure a tight seal against the finished floor surface. Coordination with the flooring contractor shall be required to achieve accurate alignment and proper functionality of the system. The purchaser shall make available the required electrical power supply up to an isolator located near the proposed door location. From this point, the supplier shall be responsible for all the power,control wiring as required for the control panel , subsequent distribution to all motors, starters, and associated operational components. This shall include the supply and installation of all necessary power and control cabling, earthing systems, and accessories required to ensure the safe and reliable operation of the door system.The supplier shall ensure that the complete installation complies with all applicable national and international standards, including relevant Indian Standards (IS) and recognized safety regulations, and is suitable for continuous operation in an aircraft hangar environment.

The hangar door system will consist of 3 mullions and 4 leaves to cover a clear width of 67.43 m, as detailed below:

2 leaves at both ends with a width of 13.00 m to 14.00 m and a height of minimum 11.50 m

2 leaves at the centre with a central mullion, having an overall width of 39.50 m to 41.50 m and a minimum height of 14.00 m.

The structural support for the hangar door shall be provided on the RCC corbel offset by 0.35 m from the end grid

As the eave height of the building is designed as 14.00 m, the door height including stacking height shall be restricted to 14.00 m for the leaves at both ends.

Codes and Standards

The minimum standards for products specified in this section shall be including as under but not limited to the following. Except as otherwise specified herein, perform work in accordance with specifications, codes and standards cited therein, and their latest editions, revisions, applicable addenda, and supplements. Where there is conflict between the referenced standards the most stringent of the conditions/requirements shall be applicable.

Signature & Seal of the Bidder)

European Standards (EN)

a. EN 12424 — Industrial, commercial and garage doors. Thermal transmittance. Requirements for the calculation

b. EN 12425 — Industrial, commercial and garage doors and gates. Resistance to water penetration. Classification

c. EN 12426 — Industrial, commercial and garage doors and gates. Air permeability. Classification

d. EN 12453 — Industrial, commercial and garage doors and gate - Safety in use of power operated doors – Requirements

e. EN 12428 — Industrial, commercial and garage doors. Thermal transmittance. Requirements for the calculation

f. EN 13501-1 — Industrial, commercial and garage doors and gates. Product standard. Reaction to Fire characteristics.

g. EN 60204-1 — Safety of machinery. Electrical equipment of machines.

h. EN 13241-1 — Industrial Standard including all normative references.

i. CEN conformity

j. TUV Certification (Certification of Safety Arrestors)

k. DIN, BS and ASTM Norms

Submittals

Product Data: Manufacturer's data sheets on each product to be used, including:

(a) Preparation instructions and recommendations.

(b) Storage and handling requirements and recommendations.

(c) Installation methods.

Shop Drawings: Include opening dimensions and door characteristics including size, clearances, and load diagrams; Suitable interface details for jamb, head, seal, and connections, hardware, safety devices, and mechanical device descriptions including motor sizes in KW, voltage phase, and hertz; location of control panel and drive units; and all design and detail data for work of other trades affected by the installation of the overhead hoist-up fabric door.

Samples: Submit colour samples of panel fabric for approval.

Operation and Maintenance Data: Submit three (3) copies of the manufacturer's operation and maintenance manuals including parts lists, as-built drawings along with Part Identification lists (PIL) & Manufacturers recommended list of spares, information on any other specific feature/s information needed for proper operation and maintenance of Vertical Lifting Fabric doors.

COCHIN INTERNATIONAL AVIATION SERVICES LIMITED * PIN 683 111

Signature & Seal of the Bidder)

Quality Assurance

The material, construction, and operation of the door shall conform to the performance specifications contained herein.

Design Requirements

Door Design: The manufacturer shall design the door in accordance with the criteria specified herein. Doors shall operate properly without binding, interference, or damage to the adjacent structure. Colour/Insignia/Text details shall be discussed and finalized with the Engineer in Charge after awarding of the work.

Wind Load: Door sections shall be designed to withstand the wind speed of 180kmph.

Door Speed: Door shall open fully in less than 162 seconds under all design conditions.

Door Weight: The door manufacturer shall provide load diagrams of the doors in the closed and open positions. Details shall also be provided showing clearance and attachment requirements for coordination with the structural steel and miscellaneous steel shop drawings.

The door must be designed to allow a maximum creep of 50 mm to make sure to avoid heavy loads on the adjacent structure. The Supplier must ensure accurate design, manufacture, supply, installation, and commissioning in respect of the proper intended use and functioning of the Vertical Lifting Fabric Door.

Delivery, Storage, and Handling

Delivery of Materials: Manufactured materials shall be delivered in original rolls, packages, boxes, or crates bearing the manufacturer's name, brand, model number, and installation and with identification labels intact Use care in handling to prevent damage.

Storage: All materials shall be carefully stored in an area that is protected from deleterious elements. Storage shall be in a manner that will prevent damage or marring of the materials and its finish, properly protected from construction activities and other possibilities of damage or loss.

Project Conditions

Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by the manufacturer for optimum results. Do not install products under environmental conditions outside the manufacturer's absolute limits.

Warranty and Maintenance

Two-year warranty on any defects related to materials and workmanship. If a defect arises within 24 months of installation which is attributable to faulty design, materials, and workmanship, the same should repaired/replaced without any additional charges to customer.

Door Leaf: The door leaf shall be made of two layers of very strongly vinyl coated polyester fabric weighing approximately 900g/m2 Capable of carrying 4000 N/5cm.It shall be UV stabilized, self-extinguishing (0 -25 flame spread), and suitable to withstand temperatures in the range between – 35°C to +55°C. The fabric shall be attached to both sides of the intermediate beams, top beam, and bottom beam with self-tapping screws through corrosion-resistant

Signature & Seal of the Bidder)

INTERNATIONAL AVIATION SERVICES PIN 683111

aluminium batten strips Powder coated clamp strips to match the color of the fabric. The door fabric shall be pulled tight between the intermediate beams when the door is fully closed, and the wind lock engaged. The fabric shall have been tested to meet the minimum criteria of the attached test documents and standards as below:

(a) **Flame retardancy:** tested according: DIN 4102:B1,EN 13501-1, B-s1,d0.

(b) **Tensile strength:** 2500/2000 N / 5 cm acc to EN ISO 1421-1

(c) **Tear resistance:** 250 / 250 N acc to DIN 53363

(d) **Light resistance:** ≥ 7 acc to EN ISO 105-B02

(e) **Radar reflecting coefficient:** 0,3dB, ~0,1%

(f) **Crack resistance:** No cracks, DIN 53359A

(g) **Material :**PES DIN EN ISO 2076

(h) **Yarn Count:** 1100 to 1670 dtex polyster DIN EN ISO 2060

Door Color: Color shall be discussed and finalized with the Engineer in Charge after awarding of the work.

Doors shall consist of hoist-up fabric doors with intermediate aluminum beams. Fabric shall be gathered above the head of the opening. Regardless of location, the minimum design wind load shall be only calculated to EN code provision and standards.

Door Beams: The intermediate beams shall be corrosion protected and they shall be spaced 1.2 to 1.5 meters apart, dependent upon the wind load. At each end of the beam, there shall be a guide block of self-lubricating nylon material that runs along the guides. The intermediate beams must have a minimum depth of 390 mm. The intermediate door beams and guide block shall be designed to carry the full design wind load without failure or being pulled from the door guides. Intermediate beams shall be made of extruded aluminium and there shall be no slots/holes made in the intermediate beams for lifting belt or ropes to pass. Steel intermediate beams are not acceptable.

Bottom Section: The bottom section is to be made of steel. A rubber seal to be fixed to the section to ensure tightness against the floor /ground. The bottom section should prevent the lifting of the door during wind load and be secured a locked position. The bottom beam shall be designed with suitable width and depth to carry the load of the intermediate beams when the door is open and to ensure full closing and a tight floor seal in heavy winds.

Door Guide: The vertical guides shall be made of aluminium only. The guides shall be designed to provide weather sealing on the inside and the outside faces. The guides shall be designed to provide sufficient strength to transfer the wind load and safety arrestor forces from the door beams, at maximum wind load. The contractor shall furnish jamb structural steel fasteners of the size and type required for the attachment of the guide rails as shown on the approved shop drawings.

COCHIN INTERNATIONAL AVIATION SERVICES LIMITED * PIN 683111

Signature & Seal of the Bidder)

Drive Units: Each door leaf shall have an EN labelled motor drive system. The gear motor shall be equipped with a drum on which the belt/ropes are wound. Only a single motor drive is to be used and the belt/ropes shall be wound on the same drum. The belt/ropes shall be attached to the bottom beam. The gear motor shall be removable without disturbing the limit switch adjustment. Each door leaf shall have only one drive motor, multiple motors with a shaft drive system is not acceptable.

Belt/Rope System: A maximum of two belts/ropes per door shall be used to transmit motive force to the door unit. Similarly, the quantity of sheaves used to guide the belt/ropes shall be minimized to reduce maintenance requirements and spare parts inventories. Belts shall be installed free of any kinks and the system design and sheave diameter shall be such to prevent the occurrence of any kinks or abnormal stress in the operating belt/ropes. Where belts pass through openings in the building structure the openings shall be constructed to prevent abrasion, wear or damage to the belts/ropes. Sheave units shall be installed in accessible locations that allow inspection and preventive maintenance. Sheaves shall not be in enclosed locations that are not readily accessible for visual inspection. Load testing of the belts needs to be undertaken and certification for the same needs to be submitted to the Employer. Multiple belts/ropes drive system is not acceptable.

Control Panel: The door manufacturer shall supply touch screen PLC-based control panels/ to operate the door units. The control panel shall also contain interlocks to preclude personnel injury, including an interlock between the power supply system and the use of the hand crank for the manual operation of the door unit. The control panel shall be mounted a minimum of 1200mm above the floor. The control panel shall be IP 55 rated.

Self-Activating Wind Lock: Each door leaf shall be provided with two self-activating wind locks which activate and lock the bottom beam when the door reaches its closed position. This locking action shall act to maintain a tight floor seal and intermediate stability even under conditions of heavy wind. A proximity switch, with an indicator, shall be provided to notify the door operator that all wind locks are engaged properly. A third-party certified Wind lock/Safety arrestor is a mandatory requirement for the door system.

Wind Resistance: Doors shall be engineered to take into account the local environmental conditions i.e., high winds, high temperatures, high levels of dust, and high humidity levels. Door sections shall be designed to withstand wind loads of 180kmph.

Water penetration: Doors shall be engineered in accordance with EN12425 Class 3 with resistance to water penetration of 110PA for a closed door with or without a pass door. The door shall be exceptionally designed to withstand water spray for > 20 minutes under the pressure of >50PA (N/m^2).

Air Tightness: Doors shall be engineered in accordance with EN 12426 **Class 2- 3,** with or without pass-door. The air permeability of the door shall be 6-12$m^3/m^2/h$ at a pressure of 50PA.

Electric Power Operations

(a) **Motors**

(i) Doors shall be provided with electrically operated motors and controls wired for operation on voltage indicated on the electrical drawings.

Signature & Seal of the Bidder)

(ii) Motors shall be the heavy-duty industrial type with IP55 protection class and only one motor to be used for each door leaf.

(iii) The power supply shall be 03 phase with 415V/50Hz input supply, **or other if specified.**

(b) Controller

(i) Doors shall be supplied with PLC-based touch screen control with protection class IP 55. unit. The control unit shall command the gear motor via buttons on the touch screen.

(ii) Fire alarm integration: Vertical lifting fabric doors shall be provided with an additional provision of FA integration. Automatic opening of door in case the fire alarm gives a signal.

(iii) Diagnostic program that will alarm the faults.

(iv) The control panel shall be wired to accept a furnished plug-in generator to operate doors in case of a power failure.

(c) **Electrical Operation:** Audible and visual alarm will activate five seconds prior to door movement and during door movement. When the door is completely opened it shall stop on the primary top limit switch. In case of over travel, a secondary limit switch shall cut off the drive unit's motor current to prevent damage. These two limit switches shall be activated by the uppermost intermediate beam. The drive unit shall be stopped by the slack cable breakers when the door is closed. The slack breakers shall also stop the door in case of belt/rope rupture or if an obstruction should prevent the door from being closed. The slack breakers shall be activated by weight or springs.

(d) **Auxiliary Emergency Operation:** The control panel shall be fed from a transfer switch that will supply emergency power to the door systems in case of a power failure. The control panel shall have a set of 415V terminals which can be connected to a dry contact located in the customer's transfer switch. The contact will be open when the transfer switch is in the normal and center position. The contact will close when the transfer switch is in the emergency power position. When in the emergency power position door operation will be one-door-at-a-time operation to prevent emergency power overload. The wiring from the control panel to the automatic transfer switch shall be by the Electrical Section.

Opening time: The full opening time of each door shall be a maximum 162 Sec

COCHIN INTERNATIONAL AVIATION SERVICES LIMITED
PIN
683111

Signature & Seal of the Bidder)

EXECUTION OF WORK ON SITE

Protective Coatings

Cleaning: After fabrication, clean all metal surfaces thoroughly of all mill scale, rust, oil, grease, and other foreign substances

Shop Painting: After cleaning, coat with primer all steel surfaces other than machine-finished parts. Keep paint off finished bearing surfaces. Before assembly, prime surfaces will be inaccessible after assembly. Handle painted material with care to avoid scraping or breaking the protective film.

Aluminum intermediate beams and guide rail shall be provided in 6000 series corrosion-resistant extruded aluminium with a mill finish.

Clean doors after erection on both interior and exterior.

Remove labels and visible markings.

Painting Specification and System shall conform to ISO: 12944 Part 6 classification C3.

Erection

Do not erect door until the work of other trades in preparing the opening has been completed and the roof is completed and under full dead load. The structural steel erection sub-contractor shall adjust the collateral steel for the door to the proper line, gage, and elevation in accordance with the approved tolerances on the drawings. Any discrepancies in the prepared opening shall be brought to the attention of the General Contractor before beginning the door installation.

Building Power: If the electrical power to install, test, and adjust the door under power. Power required shall be in the scope of successful contractor only.

All conduits, wire, junction boxes, accessories, and all labour to wire and connect to and between all electrical equipment on or for the door shall be installed in accordance with the door manufacturer's approved wiring diagrams and the Electrical Section of this specification. The electrical control equipment will be furnished by the door manufacturer. All conduits, wire, accessories, fused disconnect switches, and labour will be supplied by the Contractor. The electrical wiring of the door shall be coordinated with the installer during installation.

Assembly: Assemble and install the doors and accessories in accordance with approved shop drawings. After erection is complete and before field painting is applied, thoroughly clean all abraded surfaces, field welds, and field bolts, coat with primer paint.

Cleaning: Clean doors after erection on both interior and exterior.

Testing: Upon completion of the door installation, the door manufacturer or his representative shall perform a complete operating test in the presence of the Client/ Consultant. Correct all defects disclosed by the test. Re-test the doors and adjust until the entire installation is fully operable and acceptable. Various tests/ their schedule and scope of the tests will be submitted to the Engineer-in-Charge and accordingly Certificate will also be submitted during different stages of the Fabrication, Installation and commissioning as required.

COCHIN INTERNATIONAL AVIATION SERVICES LIMITED * PIN 683 111

Signature & Seal of the Bidder)

TRAINING AND REPAIR

A door fabric patch kit shall be supplied with approximately 3 sqm of fabric and all other materials required for door panel repair.

A training session shall be provided to minimum 10 persons for the door operating personnel and maintenance department. The training shall outline door operation, troubleshooting, and repair guidelines.

LIST OF APPROVED MAKE

1	Structural Steel -MS Tubular and Built up sections.	Tata, Vizag, SAIL
2	Fabric Door	Gandhi Automation Pvt. Ltd, Avians Innovations Technology Pvt Ltd, NIHVA Technologies Pvt Ltd /ASSAABLOY Mega Door Technologies Pvt Ltd or equivalent as approved

COCHIN INTERNATIONAL AVIATION SERVICES LIMITED PIN 683 111

Signature & Seal of the Bidder)